

A Self-Contained Resolution of the Integral Equation

$$y(x^3 - z^2) = x - 1$$

Abstract

We give a completely elementary proof that the Diophantine equation

$$y(x^3 - z^2) = x - 1$$

has infinitely many integer solutions. In fact, we prove a precise description of the fibre above $x = 1$, isolate the exceptional case $x^3 = z^2$, and record the exact divisibility criterion governing every remaining solution. The infinite families obtained include infinitely many solutions with all three coordinates non-zero.

1 Statement of the problem

We study integer triples

$$(x, y, z) \in \mathbb{Z}^3$$

satisfying

$$y(x^3 - z^2) = x - 1. \tag{1}$$

The goal is not to solve a finite search problem, but to decide whether there are infinitely many integer solutions. The answer is affirmative, and the proof is short but complete.

2 Elementary preliminary facts

We use only the following elementary properties of the ring $\mathbb{Z}$.

Lemma 1 (Zero-product principle in $\mathbb{Z}$). *If $a, b \in \mathbb{Z}$ and $ab = 0$, then $a = 0$ or $b = 0$.*

Proof. Suppose $a \neq 0$. If also $b \neq 0$, then $|a| \geq 1$ and $|b| \geq 1$, so

$$|ab| = |a| |b| \geq 1,$$

which contradicts $ab = 0$. Hence, if $ab = 0$, at least one of a, b must be zero. □

Lemma 2 (Integer square roots of 1). *If $z \in \mathbb{Z}$ and $z^2 = 1$, then $z = 1$ or $z = -1$.*

Proof. The equality $z^2 = 1$ is equivalent to

$$(z - 1)(z + 1) = 0.$$

By Lemma 1, either $z - 1 = 0$ or $z + 1 = 0$. Thus $z = 1$ or $z = -1$. □

3 The fibre above $x = 1$

The key observation is that the right-hand side of (1) vanishes when $x = 1$. This immediately produces infinite families, but we first state the exact fibre.

Proposition 1 (Exact solution set with $x = 1$). *The integer solutions of (1) with $x = 1$ are precisely the triples belonging to*

$$\{(1, 0, t) : t \in \mathbb{Z}\} \cup \{(1, t, 1) : t \in \mathbb{Z}\} \cup \{(1, t, -1) : t \in \mathbb{Z}\}.$$

Proof. Substituting $x = 1$ into (1) gives

$$y(1^3 - z^2) = 1 - 1,$$

so

$$y(1 - z^2) = 0.$$

By Lemma 1, either $y = 0$ or $1 - z^2 = 0$.

If $y = 0$, then z is arbitrary, and we obtain exactly the triples

$$(1, 0, t), \quad t \in \mathbb{Z}.$$

If $1 - z^2 = 0$, then $z^2 = 1$, and Lemma 2 gives $z = 1$ or $z = -1$. In either case y is arbitrary. Hence we obtain exactly the triples

$$(1, t, 1), \quad t \in \mathbb{Z},$$

and

$$(1, t, -1), \quad t \in \mathbb{Z}.$$

Conversely, every triple in these three displayed families satisfies $x = 1$ and either $y = 0$ or $z^2 = 1$. Therefore

$$y(1 - z^2) = 0 = 1 - 1,$$

so each such triple satisfies (1). This proves the claimed exact description. $\square$

4 The exceptional case $x^3 = z^2$

Although the proof of infinitude already follows from Proposition 1, it is useful to separate the only case in which the factor $x^3 - z^2$ vanishes.

Lemma 3 (Vanishing of the cubic-square factor). *Let $(x, y, z) \in \mathbb{Z}^3$ satisfy (1). If*

$$x^3 - z^2 = 0,$$

then

$$x = 1 \quad \text{and} \quad z \in \{1, -1\}.$$

Conversely, every triple $(1, y, 1)$ and every triple $(1, y, -1)$, with $y \in \mathbb{Z}$, satisfies (1) and has $x^3 - z^2 = 0$.

Proof. Assume (x, y, z) satisfies (1) and $x^3 - z^2 = 0$. Then the left-hand side of (1) is

$$y(x^3 - z^2) = y \cdot 0 = 0.$$

Therefore the right-hand side must also be zero:

$$x - 1 = 0.$$

Thus $x = 1$. Substituting this into $x^3 - z^2 = 0$ gives

$$1 - z^2 = 0,$$

so $z^2 = 1$. By Lemma 2, $z = 1$ or $z = -1$.

Conversely, if $(x, z) = (1, 1)$ or $(x, z) = (1, -1)$, then

$$x^3 - z^2 = 1 - 1 = 0$$

and

$$x - 1 = 0.$$

Thus, for every $y \in \mathbb{Z}$, both sides of (1) are zero. Hence all such triples are solutions. $\square$

5 Divisibility criterion for the remaining cases

The equation is linear in y . Therefore, once x and z are chosen with $x^3 \neq z^2$, the value of y , if it exists, is forced.

Proposition 2 (Exact divisibility criterion away from $x^3 = z^2$). *Let $x, z \in \mathbb{Z}$ satisfy*

$$x^3 - z^2 \neq 0.$$

Then there exists an integer y such that (x, y, z) satisfies (1) if and only if

$$x^3 - z^2 \mid x - 1.$$

In that case the only possible value of y is

$$y = \frac{x - 1}{x^3 - z^2}.$$

Proof. Set

$$D = x^3 - z^2.$$

By hypothesis, $D \neq 0$. Equation (1) becomes

$$yD = x - 1.$$

First suppose that an integer y satisfying this equation exists. Then $x - 1$ is an integer multiple of D , namely

$$x - 1 = yD.$$

Thus $D \mid x - 1$, which is exactly

$$x^3 - z^2 \mid x - 1.$$

Moreover, since $D \neq 0$, division by D gives the forced value

$$y = \frac{x - 1}{D} = \frac{x - 1}{x^3 - z^2}.$$

Conversely, suppose $D \mid x - 1$. By definition of divisibility in $\mathbb{Z}$, there is an integer q such that

$$x - 1 = qD.$$

Taking $y = q$, we get

$$y(x^3 - z^2) = qD = x - 1,$$

so (x, y, z) satisfies (1). The displayed formula for y follows again from $D \neq 0$. $\square$

6 Infinitely many integer solutions

We now give the promised unconditional conclusion.

Theorem 1 (Infinitude of integer solutions). *The Diophantine equation*

$$y(x^3 - z^2) = x - 1$$

has infinitely many integer solutions.

Proof. For each integer n , define

$$(x_n, y_n, z_n) = (1, n, 1).$$

Then

$$y_n(x_n^3 - z_n^2) = n(1^3 - 1^2) = n(1 - 1) = n \cdot 0 = 0,$$

while

$$x_n - 1 = 1 - 1 = 0.$$

Hence

$$y_n(x_n^3 - z_n^2) = x_n - 1,$$

so $(1, n, 1)$ is an integer solution for every $n \in \mathbb{Z}$.

These solutions are pairwise distinct: if $(1, n, 1) = (1, m, 1)$, then equality of the second coordinates gives $n = m$. Since $\mathbb{Z}$ is infinite, the set

$$\{(1, n, 1) : n \in \mathbb{Z}\}$$

is infinite. Therefore (1) has infinitely many integer solutions. $\square$

Corollary 1 (Infinitely many non-zero-coordinate solutions). *There are infinitely many integer solutions (x, y, z) of (1) with*

$$xyz \neq 0.$$

Proof. For every non-zero integer n , the triple

$$(1, n, 1)$$

is a solution by the proof of Theorem 1. Its coordinate product is

$$1 \cdot n \cdot 1 = n \neq 0.$$

Since there are infinitely many non-zero integers n , this gives infinitely many solutions with all three coordinates non-zero. $\square$

7 Summary of the solution set information proved here

Combining the preceding results, we have the following complete structural statement.

Theorem 2 (Structural description). *An integer triple $(x, y, z) \in \mathbb{Z}^3$ satisfies*

$$y(x^3 - z^2) = x - 1$$

if and only if exactly one of the following alternatives holds:

- (i) $x = 1$, $z \in \{1, -1\}$, and $y \in \mathbb{Z}$ is arbitrary;
- (ii) $x = 1$, $z \notin \{1, -1\}$, and $y = 0$;

(iii) $x \neq 1$, $x^3 - z^2 \neq 0$, $x^3 - z^2 \mid x - 1$, and

$$y = \frac{x - 1}{x^3 - z^2}.$$

In particular, alternatives (i) and (ii) already contain infinite families of integer solutions.

Proof. Let $(x, y, z) \in \mathbb{Z}^3$.

Assume first that (x, y, z) is a solution. If $x = 1$, Proposition 1 says precisely that either $z \in \{1, -1\}$, with y arbitrary, or else $z \notin \{1, -1\}$, in which case the only remaining possibility is $y = 0$. Thus a solution with $x = 1$ falls into exactly one of alternatives (i) and (ii).

Now suppose $x \neq 1$. A solution cannot have $x^3 - z^2 = 0$, because Lemma 3 would then force $x = 1$. Hence $x^3 - z^2 \neq 0$, and Proposition 2 gives exactly the divisibility condition and the displayed formula for y . Thus the solution falls into alternative (iii).

Conversely, every triple in alternative (i) or (ii) is a solution by Proposition 1, and every triple in alternative (iii) is a solution by Proposition 2. The three alternatives are disjoint because alternatives (i) and (ii) require $x = 1$, alternative (iii) requires $x \neq 1$, and alternatives (i) and (ii) impose mutually exclusive conditions on z . This proves the stated structural description. $\square$